

Zeng Shengjie

Email: 24102082d@connect.polyu.hk | Phone: 59579451 | Student ID: 24102082D

Education Background

PolyU, Bachelor of Science in Computer Science

Hong Kong, China | 2024.08-2028.08

Major Courses: Data Structures and Algorithms, Operating Systems, Computer Networks, Database Systems, Introduction to Artificial Intelligence, Software Engineering, etc.;

Minor: Applied Mathematics;

Professional Competence

Programming Languages: C++, Python, Java, Rust (Proficient); Familiar with modern C++ (17/20) and Rust async programming;

AI/ML: PyTorch, Scikit-learn; Spatio-temporal Transformer, GNN/GAT; LightGBM/XGBoost/CatBoost ensemble learning; Optuna hyper-parameter tuning; SHAP, PyTorch-Hook attention visualization for explainable AI; spatio-temporal multi-modal fusion, spatio-temporal anomaly detection, ablation study design

System and Underlying Layer: multithreaded concurrency, Mutex/Condvar, deadlock detection; processor microarchitecture, cache coherence, SIMD, memory-access optimization

Tools and Platforms: Git, Linux Bash, OpenCV (cv2), Python aiohttp, DirectShow; FastAPI, Streamlit, AWS EC2

Internship Experience

Amazon AWS Smart City Solutions Group | Machine Learning R&D Intern

Beijing, China | 2026.06-2026.08

- **Model R&D and Optimization:** Responsible for the R&D of the urban spatiotemporal crowd-flow prediction and anomaly detection system. Implemented and compared three spatiotemporal deep-learning models: STF-Transformer, AGFormer and STGCN. Adopted a lightweight spatiotemporal decoupling architecture to compress the parameter volume to 1/10 of that of AGFormer, with the MAE on the test set reduced by 50.2%.
- **Feature Fusion and Detection:** Designed a static-dynamic conditional cross-attention multimodal fusion scheme to integrate four categories of features: POI, road network, meteorology and temporal periodicity. Built a four-paradigm anomaly-detection fusion framework combining statistical threshold, prediction residual and autoencoder. Strictly separated validation and test sets to prevent data leakage, achieving a final detection F1-score of 0.78.
- **Deployment and Interpretability:** Completed model service deployment based on AWS EC2 (g4dn.xlarge), and built FastAPI interfaces together with the Streamlit visualization system. Optimized via LRU cache and orjson serialization, the cache-hit response latency was reduced to 9.8 ms and the system throughput reached 337 steps per second, satisfying real-time early-warning requirements. Introduced SHAP feature attribution and attention heatmap interpretability analysis to quantify the contribution of spatiotemporal features to prediction outputs. Delivered comprehensive technical reports and visualization outputs, and the solution was adopted by the team.
- **Project code and full documentation:** <https://github.com/ZENGShengjie/spatio-temporal-data-pipeline.git>

Research Experience

Remote Visual-Force Feedback Teleoperation System for Robot Hair-Cutting Hong Kong, China | 2026.03-2026.06

Scenario, Team-7 Members, Person in Charge of Robotic Arm Control Module

- **Visual Perception and Calibration:** Build a USB binocular-camera depth-sensing system, complete ChArUco binocular calibration and epipolar rectification with re-projection error RMS < 1px; locate and fix the core issue of mismatch between calibration dimension and runtime resolution, achieve end-to-end native 1080p operation, and reduce systematic error by 60%.
- **Algorithm Optimization and Noise Reduction:** Design a single-channel detection + epipolar-geometry ROI mapping scheme to eliminate YOLO inference on the right channel, cutting the computational overhead for object detection by 50%; implement a four-layer depth-noise-reduction pipeline (histogram peak value + trimmed mean + Gaussian spatial weighting + confidence-adaptive EMA), lowering numerical fluctuation by 70%.
- **System Performance Improvement:** Optimize the camera reading pipeline and decoding buffer, raising the frame rate from 1fps to 6-8fps. Within the 10-30 cm ranging range, the average error is ≤ 2 cm, satisfying the real-time accuracy requirements for the robot hair-cutting scenario.

Project Experience

Medical Care Robot Coordination System, Team-4 Members, Core Member

Hong Kong, China | 2026.03-2026.05

- **Concurrency Control:** Implement a priority-preemptive task queue based on Rust (PR + FIFO + preempt). `add_task_with_preempt` enables high-priority tasks to jump the queue directly. Regional mutual-exclusion locks are implemented with `Mutex<HashMap>`, enforcing forced offline status upon a 10-second waiting timeout.
- **Health Monitoring and Self-Healing:** A dedicated thread monitors robot heartbeats (marked as dead after a 3-second timeout). Simulate deadlock/livelock scenarios, with a 12-second threshold to forcibly terminate livelocked instances. `auto_heal_system` dynamically replenishes threads to maintain the target thread count.
- **Deadlock Detection:** Implement the wait-for graph and cycle detection algorithm. Low-priority robots are preferentially sacrificed to force resource release, and incomplete tasks are re-enqueued. Condvar is used instead of busy-waiting to significantly reduce CPU overhead.
- **Performance Results:** Zero region overlaps under stress tests. A throughput of 94.8 tasks/s is achieved with 5 robots processing 1000 tasks; 41.5 tasks/s under high-contention scenarios with 10 robots; mixed-load processing of 150 tasks takes only 0.83 seconds. All concurrency and preemption tests are passed.

Kaggle Diabetes Prediction Challenge, Team-3 Members, Model Design Hong Kong, China|2026.03-2026.04

- **Feature Engineering:** Clean large-scale data, construct blood lipid ratio, blood pressure derived features and comprehensive risk scores; introduce K-Means patient clustering and PCA dimensionality reduction to generate unsupervised augmented features;
- **Model Ensemble:** Implement the ensemble of LightGBM, XGBoost, CatBoost and Ridge regression, and dynamically optimize weights using the bagging hill-climbing algorithm; configure 5-fold stratified cross-validation and early-stopping strategy, and enable GPU-accelerated training;
- **Achievements:** Public AUC of 0.70137 and private AUC of 0.69817; design a modular end-to-end framework documenting seven iterations, which has been adopted by the team;

Campus Experience

Core member of PolyU Swimming Team, National Level-1 Athlete Hong Kong, China|2024.09-now

- Represented PolyU at the Hong Kong University Sports Association Swimming Competition, winning the Men's Overall Championship in 2024 and the Overall Third-Place in 2025;
- Ten years of competitive swimming training, with excellent capabilities including stress resistance, self-discipline and concentration;

Achievement

- The Most Outstanding Academic Achievement Award of The Hong Kong Polytechnic University, 2025-2026;
- Talent Development Award of the HKSAR Government Scholarship Fund, 2024-2025 Academic Year;
- Outstanding Student-Athlete Award of The Hong Kong Polytechnic University, 2024-2025 Academic Year;
- Men's Overall Championship, Hong Kong University Sports Association Swimming Competition 2024;

Skill

Languages: Chinese (native), English (proficient, working- language proficiency);
Certificate: National Level-1 Swimmer;
Hobbies: competitive swimming, research on AI inference engines, low-level performance optimization, an programming development.